# QUANTILE SECTORAL VOLATILITY SPILLOVERS

## The Case of Vietnam and US Stock Markets

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## Introduction

**Problem:**
Rising global integration and extreme shocks → need of systemic risk assessment
**Research Objectives:**

- How do sectoral return spillovers behave across asymmetric market conditions and different time frequencies?
- Which sectors act as systemic Transmitters vs. Receivers?
- How do Vietnam's risk dynamics differ from a developed benchmark ?

## Literature Review

**Why Vietnam ?**

- One of 30 representative emerging markets by IMF (Psalida & Sun, 2009)
- Sectoral concentration enables systemic risk mapping (Bekaert & Harvey, 1997)

**Why USA ?**

- The highest liquidity, size, institutional accessibility (MSCI, 2023)
- Longest reliable historical series for long-horizon financial analysis (Shiller, 2015)

**Volatility Spillover Frameworks Evolution**

- **VAR-based Spillover Index** (Diebold & Yilmaz, 2012)
- **Frequency Domain Connectedness** (Baruník & Křehlík, 2018)
- **Quantile VAR** (Ando et al. 2022 / Chatziantoniou et al. 2021)
- **Quantile-Frequency Connectedness** (Chatziantoniou et al. 2022)

**Empirical Review**

- **USA (Developed Market)** — *S&P 500 sector indices From 2015 to 2024, using QVAR* (Kim et al., 2025)
- **Emerging Markets** — *Countries Aggregate Indices From 2003 to 2022, QVAR - Frequency* (Dang et al., 2024)
- **Vietnam** — *Vo et al. (2024) analyzed 24 sectors (2017 - 2022), using VAR*
- **Our Contribution** — *Using QVAR-frequency simultaneously for Vietnam and USA*; Update data to 2025

## Methods

### Data

- 10 sectors in the Vietnamese stock market (Jan 6, 2016 – Oct 1, 2025)
- 9 sectors in the U.S. stock market (Jan 5, 2010 – Oct 30, 2025).

### Model

**Univariate Volatility**
GJR – GARCH (1,1)

$$h_t = \omega + \sum(\alpha_k + \gamma_k I_{t-k})\varepsilon_{t-k}^2 + \sum \beta_j h_{t-j}$$

**Multivariate Correlation**
DCC –GARCH

$$Q_t = (1 - a - b)\bar{Q} + a(\eta_{t-1}\eta_{t-1}') + bQ_{t-1}$$

**Connectedness**
TCI QVAR & Frequency

$$TCI_\tau = \frac{\sum \Phi_{ij,\tau}(H)}{\sum \Phi_{ij,\tau}(H)} \times 100$$

**Portfolio Analysis**
Optimal weight

$$W_t = \frac{H_t^{-1}1}{1'H_t^{-1}1}$$

## Descriptive Statistics

Data in log-returns. All series pass Stationary tests (ADF/KPSS) and exhibit significant ARCH effects, validating the transition to volatility spillover analysis

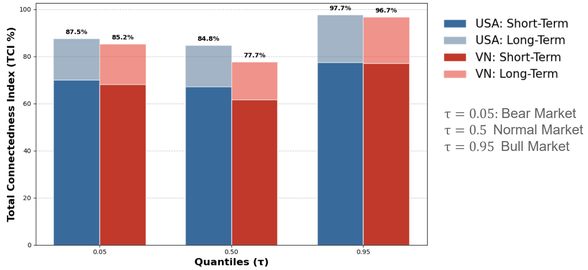
| Vietnam | Banks | Financials | Oil Gas | Basic Mat. | Industrials | Technology | Utilities | Health Care | Consumer Services | Consumer Goods |
|---|---|---|---|---|---|---|---|---|---|---|
| Mean | 0.0008 | 0.0007 | 0.0002 | 0.0007 | 0.0007 | 0.0009 | 0.0007 | 0.0006 | 0.0007 | 0.0004 |
| Std | 0.0144 | 0.0123 | 0.0192 | 0.0137 | 0.0094 | 0.0139 | 0.0129 | 0.0087 | 0.0126 | 0.0092 |
| Min | -0.0507 | -0.0415 | -0.0621 | -0.0477 | -0.0318 | -0.0453 | -0.0422 | -0.0260 | -0.0408 | -0.0314 |
| Max | 0.0421 | 0.0336 | 0.0535 | 0.0349 | 0.0242 | 0.0427 | 0.0375 | 0.0282 | 0.0354 | 0.0230 |
| Skew | -0.5021 | -0.4505 | -0.2912 | -0.6666 | -0.5920 | -0.1038 | -0.2584 | 0.0789 | -0.3313 | -0.5789 |
| Kurt | 5.3847 | 4.8278 | 4.4439 | 4.9200 | 4.6976 | 4.9369 | 4.8456 | 4.6178 | 4.7505 | 4.6308 |

| USA | Materials | Energy | Financials | Industrials | Technology | Staples | Utilities | Health Care | Discretionary |
|---|---|---|---|---|---|---|---|---|---|
| Mean | 0.0003 | 0.0003 | 0.0005 | 0.0005 | 0.0007 | 0.0004 | 0.0004 | 0.0004 | 0.0006 |
| Std | 0.0120 | 0.0153 | 0.0123 | 0.0109 | 0.0124 | 0.0076 | 0.0096 | 0.0093 | 0.0118 |
| Min | -0.0353 | -0.0458 | -0.0378 | -0.0338 | -0.0400 | -0.0230 | -0.0304 | -0.0292 | -0.0376 |
| Max | 0.0332 | 0.0424 | 0.0365 | 0.0314 | 0.0336 | 0.0202 | 0.0256 | 0.0244 | 0.0318 |
| Skew | -0.1928 | -0.1351 | -0.2318 | -0.2654 | -0.4008 | -0.2762 | -0.3404 | -0.3546 | -0.3996 |
| Kurt | 3.8152 | 3.9690 | 4.3301 | 4.1460 | 4.3536 | 3.7702 | 3.8462 | 3.9985 | 4.2676 |

## Results

### Volatility Contagion in USA and Vietnam

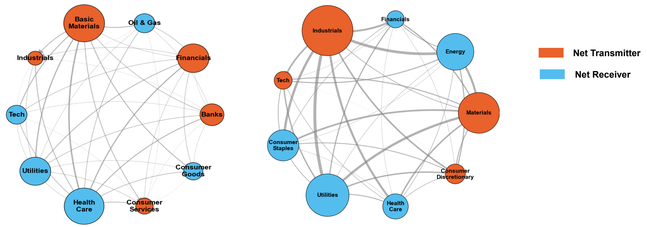
Both markets exhibit higher TCI at tail quantiles ($\tau = 0.05$ and $0.95$), primarily driven by short-term frequency shocks.




$\tau = 0.05$: Bear Market
$\tau = 0.5$ Normal Market
$\tau = 0.95$ Bull Market

### Transmitters and Receivers at Bull Market ($\tau = 0.95$)

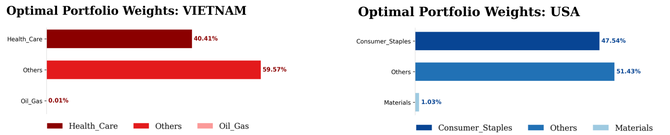
Industrials (USA) and Basic Materials (VN) act as primary risk sources, while Health Care and Utilities serve as the main risk receivers.




Vietnam (TCI: 96.7%) USA (Short-term TCI: 97.7%)

### Optimal Sector Weights for Minimum Variance Portfolio (MVP)

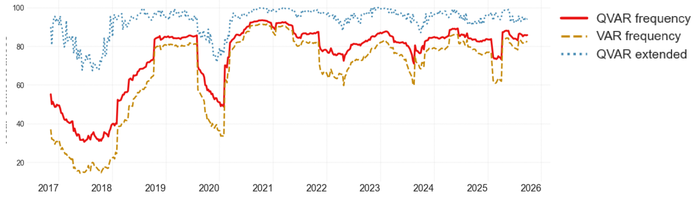
Dominance of Defensive Sectors: Health Care (VN) vs. Consumer Staples (USA).




## Robustness check

### Dynamic Total Connectedness Index in Vietnam Stock Market

Comparison across QVAR-frequency, VAR-frequency, and QVAR-extended frameworks at the median ($\tau = 0.5$)




## Discussion

- Short-term frequency components dominate total connectedness (70-80% TCI) across all quantiles, consistent with Baruník & Křehlík (2018).
- The finding of key Transmitter and Receiver in Vietnam and USA are both align with previous study:
  - Vietnam's construction-driven economy amplifies Basic Materials spillovers. (Bui et al., 2022)
  - Industrials functions as an economic bellwether and systematic risk transmitter. (Baruník & Křehlík, 2018)

## Conclusion & Policy Implications

- **Short-term Dominance:** Risk is driven by short-term shocks.
- **Procyclical Risk:** Spillovers peak during bull markets.
- **Asymmetric Integration:** Sectoral integration is uneven and heterogeneous.
- **Policy Implications:**
  - Investors: Use dynamic allocation and frequent rebalancing.
  - Risk Managers: Monitor short-term and tail-risk events.
  - Policymakers: Adopt connectedness as an early warning tool.

## References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